The object is a long-run derivative, not an annual ratio

Structural object

$$\theta_t^{\mathrm{LR}} = \left.\frac{\partial \ln Y_t^p}{\partial \ln K_t^{cap}}\right|_{\mathcal{M}^{\mathrm{LR}}}$$

Counterfactual encoded by the derivative

Increase productive capital marginally while holding the long-run distributive state and the technique inherited by the new installation fixed.

Different object

$$\vartheta_t^{path} = \frac{g_t^p}{g_t^{cap}}$$

This ratio evaluates one observed annual direction of accumulation. It does not identify the structural derivative.

Identification target

What must be estimated is the coefficient vector that governs the marginal capacity payoff of new accumulation along the long-run manifold.

Analytical source: Theta-recovery specification ledger, source line 12

Observed output mixes capacity formation with realized use

Accounting separation

$$y_t = y_t^p + \log \mu_t$$

long-run coefficient vector → $\widehat{\theta}_t^{\mathrm{LR}}$ capacity payoff → $\widehat{y}_t^p$ capacity path → level anchor → $\widehat{\mu}_t$ realized use

The residual is not the denominator

Residual stationarity admits a long-run relation. Only the reconstructed and anchored y_t^p supplies the denominator from which μ_t is calculated.

Analytical source: Distribution-conditioned identification rule, source lines 19 and following

Persistent distribution selects technique before installation

Long-run technique choice

$$q_t^* = \arg\max_q \left[a(q) - q\pi_t^{\mathrm{LR}}\right]$$

$$a'(q_t^*) = \pi_t^{\mathrm{LR}} = 1 - \omega_t^{\mathrm{LR}}$$

- q_t^* is selected using the distributive state expected to persist over the life of the installation.
- The selection occurs before the corresponding capital enters the productive stock.

Separate the horizons

$$\omega_t = \omega_t^{\mathrm{LR}} + \omega_t^{SR}$$

ω_t^{LR}	selects technique for new accumulation
ω_t^{SR}	affects realized output and utilization without repricing the installed stock

Economic implication

Contemporaneous wage-share noise cannot be allowed to make the productive capacity of yesterday's capital jump today.

Analytical source: Theta-recovery ledger, technique-choice condition at source line 63

New investment inherits yesterday's selected technique

persistent state $\omega^{\mathrm{LR}}_{t-1}$ → selected technique q^*_{t-1} → new accumulation Δk_t → embodied increment $\Delta Q^{tech}_t = \widetilde{q}^*_{t-1} \Delta k_t$

Installed stock

Capital already in place retains the technique selected when it was installed. A change in ω^{LR}_t does not revalue its physical capacity.

New layer of the stock

Only Δk_t receives the technique weight inherited from $t-1$. The capacity path therefore records the sequence of past installation choices.

Analytical source: Theta-recovery ledger, causal ordering and accumulated-path construction

Accumulation turns technique choice into a long-run state

Technique-centered accumulated path

$$\widetilde{q}_t^* = q_t^* - \bar{q}^{*,\mathrm{LR}}, \qquad Q_t^{tech} = \sum_{s \le t} \widetilde{q}_{s-1}^* \Delta k_s$$

What the state remembers

high q_{s-1}^*	accumulation installed with a technique above the long-run reference
low q_{s-1}^*	accumulation installed with a technique below the reference
no Δk_s	no new technique is embodied in that period

Centering changes the basis, not capacity

$$Q_t^{raw} = Q_t^{tech} + \bar{q}^{*,\mathrm{LR}}(k_t - k_0)$$

$$\beta_K^{centered} = \beta_K^{raw} + \beta_Q \bar{q}^{*,\mathrm{LR}}$$

Interpretation

$\beta_K^{centered}$ is the capacity elasticity at the reference technique.

Analytical source: Theta-recovery ledger, accumulated path at source line 75

The cointegrating vector recovers θ_t at every date

$$y_t = \alpha + \beta_K k_t + \beta_Q Q_t^{tech} + \varepsilon_t$$

Differentiate the reconstructed capacity path

$$\begin{aligned} \Delta y_t^p &= \beta_K \Delta k_t + \beta_Q \Delta Q_t^{tech} \\ &= \left(\beta_K + \beta_Q \widetilde{q}_{t-1}^*\right) \Delta k_t, \\ \therefore \quad \theta_t^{\mathrm{LR}} &= \beta_K + \beta_Q \widetilde{q}_{t-1}^*. \end{aligned}$$

Estimated once
α, β_K, and β_Q belong to the long-run coefficient vector.

Reconstructed each date
The lagged technique state maps the fixed vector into the historical θ_t^{LR} path.

Identification verdict: Persistence enters through the accumulated installation state, not through a sequence of unrelated annual coefficients.

Analytical source: Theta-recovery ledger, preferred recovery equation at source line 224

The profit-share formula is the reduced form of embodied technique choice

Affine long-run technique selection

$$q^*_{t-1} = a + b\pi^{\mathrm{LR}}_{t-1}$$

Substitution into the accumulated-path derivative

$$\begin{aligned}
\theta^{\mathrm{LR}}_t &= \beta_K + \beta_Q \left(q^*_{t-1} - \bar{q}^{*,\mathrm{LR}} \right) \\
&= \underbrace{\left[\beta_K + \beta_Q (a - \bar{q}^{*,\mathrm{LR}}) \right]}_{\theta_1} + \underbrace{\beta_Q b}_{\theta_2} \pi^{\mathrm{LR}}_{t-1} \\
&\equiv \theta_1 + \theta_2 \pi^{\mathrm{LR}}_{t-1}.
\end{aligned}$$

θ_1: capacity payoff at the reference long-run distributive state.

θ_2: technique-selection effect transmitted through new accumulation.

Analytical source: Chapter identification architecture, governing profit-share form at source line 567

Unity is the Harrodian benchmark

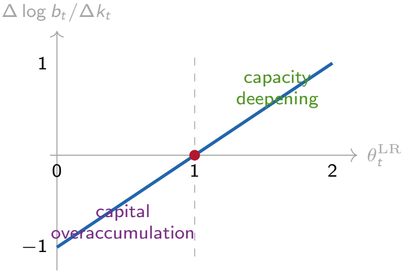
Capacity productivity

$$b_t = \frac{Y_t^p}{K_t^{cap}}$$

$$\Delta \log b_t = \left(\theta_t^{\mathrm{LR}} - 1\right) \Delta k_t$$

Conditional on $\Delta k_t > 0$:

- $\theta_t^{\mathrm{LR}} < 1$: capital grows faster than the capacity it builds;
- $\theta_t^{\mathrm{LR}} = 1$: capacity productivity remains constant;
- $\theta_t^{\mathrm{LR}} > 1$: capacity grows more than proportionally.



Admissibility

Contraction years are classified separately; they do not mechanically reverse the threshold interpretation.

Analytical source: Chapter identification architecture, Harrodian benchmark at source line 411

θ_t classifies accumulation; μ_t classifies realized use

Utilization dynamics after capacity reconstruction

$$\Delta \log \mu_t = \Delta y_t - \theta_t^{\mathrm{LR}} \Delta k_t$$

	$\mu_t < 1$: unused capacity	$\mu_t > 1$: capacity strain
$\theta_t^{\mathrm{LR}} < 1$ capital overaccumulation	**Overaccumulation with slack.** Weak capacity conversion coexists with a demand shortfall large enough to leave installed capacity unused.	**Overaccumulation with strain.** Capital accumulates faster than capacity while realized demand presses against the available envelope.
$\theta_t^{\mathrm{LR}} > 1$ capacity deepening	**Realized excess capacity.** Capacity grows more than proportionally and effective demand fails to absorb it.	**Expansion absorbed by demand.** High capacity conversion is matched by realized output, preventing unused capacity.
	$g_{\mu,t} < 0$: movement toward greater slack	$g_{\mu,t} > 0$: movement toward greater strain

The thresholds answer different questions

θ_t^{LR} identifies how accumulation builds capacity. The anchored level and movement of μ_t identify whether that capacity is used.

Analytical source: Distribution-conditioned identification sequence and explicit capacity anchoring

A regime claim requires uncertainty to clear unity

Simultaneous band	Decision rule	Structural verdict
entirely below unity	$U_t^{\theta} < 1$	below-one capital-overaccumulation regime
crosses unity	$L_t^{\theta} \leq 1 \leq U_t^{\theta}$	indeterminate; the data do not identify a side of the benchmark
entirely above unity	$L_t^{\theta} > 1$	above-one capacity-deepening regime

Structural classification also fails when

- the accumulated-path rank condition fails;
- the long-run residual is nonstationary;
- $\theta_t^{\mathrm{LR}} \leq 0$;
- cumulative capital growth is nonpositive.

Final analytical verdict

$\theta_t^{\mathrm{LR}} > 1$ establishes a tendency for capacity to outrun capital. Actual unused capacity requires $\mu_t < 1$ or a sustained $g_{\mu,t} < 0$.

Analytical source: Theta-recovery admissibility rule and generated-state inference contract

A1. Identification fails when any gate fails

Gate	Pass condition	Failure classification
state order	k_t and Q_t^{tech} are admissible long-run states	mixed-order or polynomial-risk specification
rank	$\text{rank}[k_t, Q_t^{tech}] = 2$	technique path is not separately identified from scale
residual	$\widehat{\varepsilon}_t \sim I(0)$	no stable long-run capacity relation
timing	q_{t-1}^* weights Δk_t	contemporaneous reverse-causality risk
anchoring	one explicit benchmark fixes the level of y_t^p	μ_t remains underidentified
economic range	simultaneous band remains above zero	economically inadmissible capacity payoff
accumulation	$\sum_{t \in w} \Delta k_t > 0$	no threshold interpretation for that window

Rank has an economic meaning

Technique must vary when accumulation occurs. If $\widetilde{q}_{t-1}^* \Delta k_t$ is only a constant multiple of Δk_t, the data identify scale but not a separate technique-conditioned elasticity.

Analytical source: Capacity-utilization identification rule and theta-recovery admissibility ledger

A2. Composition payoffs and annual ratios cannot classify the regime

Two-capital differential

$$dy_t^p = \theta_t^{scale}\, dk_t^{NR} + \theta_t^{\tau}\, d\tau_t, \qquad \tau_t = k_t^{ME} - k_t^{NR}$$

Object	**Counterfactual answered**	**Regime role**
θ_t^{LR}	How much does capacity change when productive capital expands along the long-run accumulation path?	structural coefficient tested against one
θ_t^{τ}	What is the capacity payoff of changing ME/NR composition at a given scale?	composition mechanism, not aggregate regime threshold
g_t^p/g_t^{cap}	What capacity payoff occurred along this particular annual direction?	realized-direction diagnostic, not an equilibrium coefficient

Balanced-accumulation verdict: Holding composition fixed gives $d\tau_t = 0$. The scale derivative remains; the composition payoff and the annual denominator do not enter the unity test.

Analytical source: Theta-recovery component-capital ledger and the long-run object distinction

A3. Generated-state uncertainty must be propagated end to end

resample time blocks → re-estimate $\omega_t^{\mathrm{LR}}, q_t^*$ → rebuild Q_t^{tech} → re-estimate β_K, β_Q → reconstruct $\theta_t^{\mathrm{LR}}, \Theta_w^{\mathrm{LR}}$

Window-level long-run elasticity

$$\Theta_w^{\mathrm{LR}} = \frac{\sum_{t\in w} \theta_t^{\mathrm{LR}} \Delta k_t}{\sum_{t\in w} \Delta k_t}, \qquad \sum_{t\in w} \Delta k_t > 0$$

Bootstrap protocol

1000 moving-block draws with block length 4 years. Every draw re-estimates the complete generated-regressor chain.

Inference output

95% simultaneous bands for θ_t^{LR} and bootstrap intervals for Θ_w^{LR}. Pointwise intervals alone do not identify sustained regimes.

Analytical source: Generated-state inference contract; 13 source claims and 9 identities validated